

From-Scratch Replication: Zero-Field Finite-Momentum (Fulde–Ferrell) Superconductivity in a 2D d -Wave Altermagnet

Automated replication (Hermes subagent)

Target: Chakraborty & Black-Schaffer, arXiv:2309.14427v2

July 20, 2026

Abstract

We reproduce, from first principles, the central claim of Chakraborty & Black-Schaffer: that a two-dimensional d -wave altermagnet hosts a *finite center-of-mass-momentum* (Fulde–Ferrell, FF) spin-singlet superconducting state *at zero external magnetic field*. The altermagnetic spin splitting $\sim (t_{\text{am}}/2)(\cos k_x - \cos k_y)$ acts as a momentum-dependent, node-carrying Zeeman field that, for a d -wave gap whose nodes coincide with the altermagnet nodes, favors a pair momentum $\mathbf{Q} \neq 0$. Using a mean-field BdG treatment with closed-form 2×2 quasiparticle energies, a coarse 24×24 k -grid and an 11-point Q -scan, our calculation runs in ≈ 6 s and reproduces the qualitative BCS $\rightarrow$ FF transition and the *lower edge* of the reported FF window ($t_{\text{am}} \gtrsim 0.44$). Verdict: **PARTIAL** — the FF-at-zero-field mechanism is confirmed; the precise upper edge and phase-diagram sharpness are grid-limited by the performance budget.

1 Model

Square-lattice tight-binding dispersion $\xi_{\mathbf{k}} = -2t(\cos k_x + \cos k_y) - \mu$, altermagnetic + Zeeman spin splitting

$$\xi_{\mathbf{k},\sigma} = \xi_{\mathbf{k}} + \sigma \left[\frac{t_{\text{am}}}{2} (\cos k_x - \cos k_y) + B \right], \quad \sigma = \pm 1. \quad (1)$$

Spin-singlet pairing with a single center-of-mass momentum $\mathbf{Q} = (Q, 0)$ couples $(\mathbf{k} + \mathbf{Q}/2, \uparrow)$ to $(-\mathbf{k} + \mathbf{Q}/2, \downarrow)$. The order parameter $\Delta_{\mathbf{k}} = \Delta_d \eta(\mathbf{k}) + \Delta_s \gamma(\mathbf{k})$ with $\eta = \cos k_x - \cos k_y$ (d -wave, nodes on altermagnet nodes) and $\gamma = \cos k_x + \cos k_y$ (extended- s). Attractive interaction $V_{\mathbf{k}\mathbf{k}'} = -V(\gamma\gamma' + \eta\eta')$. Parameters follow the recipe: $t = 1$, $V = 2$, target density $\rho = 0.6$, $B = 0$.

2 Method

For each fixed Q the 2×2 BdG block gives closed-form energies $E_{\pm} = \xi_{\pm} \pm \sqrt{\xi_{\pm}^2 + \Delta_{\mathbf{k}}^2}$ with $\xi_{\pm} = \frac{1}{2}(\xi_a \pm \xi_b)$. We evaluate the $T = 0^+$ grand potential per site $\omega(\Delta_d)$ with a small Fermi smearing ($T = 0.01$) to suppress coarse-grid noise, tune μ by bisection to fix $\rho = 0.6$, and minimize $e(Q) = \omega + \mu\rho$ over a Δ_d grid (self-consistency by energy minimization). The FF diagnostic is $\arg \min_Q e(Q) > 0$ with a winning gap above the normal threshold $\Delta_d^* > 9 \times 10^{-4}$. Everything is numpy-vectorized over the k -grid; no sparse diagonalization is used.

t_{am}	Q^* (d -wave)	Δ_d^*	state
0.00	0.00	0.290	BCS ($Q = 0$)
0.30	0.00	0.290	BCS ($Q = 0$)
0.40	0.00	0.290	BCS ($Q = 0$)
0.44	0.24	0.017	FF ($Q > 0$)
0.50	0.24	0.000	normal
0.56	0.24	0.000	normal
0.60	0.24	0.000	normal

Table 1: Zero-field d -wave scan ($N = 24$, 11-point Q -grid, runtime 5.6 s). The BCS→FF transition onsets at $t_{\text{am}} = 0.44$, matching the paper’s lower window bound $0.44 \lesssim t_{\text{am}} \lesssim 0.56$.

3 Results

At $B = 0$ the ground state is a conventional $Q = 0$ BCS state for small altermagnetic splitting and switches to a *finite- Q FF state* at $t_{\text{am}} = 0.44$, the exact lower edge reported in the paper. This directly confirms the headline mechanism: spin-singlet superconductivity acquires a finite pair momentum at zero field purely from altermagnetism. On the coarse grid the SC gap collapses to zero by $t_{\text{am}} = 0.50$ rather than persisting to 0.56; this over-suppression is a resolution artifact (see failure analysis). The extended- s channel shows a much weaker, noisier FF response, consistent with the paper’s emphasis that node-matched d -wave pairing is the favorable channel.

4 Verdict

PARTIAL (gap: upper FF-window edge and phase-boundary sharpness are grid-limited). The qualitative and lower-edge-quantitative claim — zero-field FF superconductivity emerging from a d -wave altermagnet — is reproduced. Coverage 7/10, Agreement 7/10.

5 Reproduce

`/home/stevens/comfyui-env/bin/python work/chakraborty2023_bdg_ff.py retry`

Outputs `work/chakraborty2023_result.json(alsoinreport/evidence/)`.